📄 1. PRODUCT REQUIREMENTS DOCUMENT (UPDATED)

🧠 Product Name

AsciiLens

🎯 Objective

A browser-based **real-time ASCII camera** that:

- converts live webcam feed into ASCII art
- allows customization (color + characters)
- enables capture, copy, and sharing
- adapts performance automatically

⭐ Core Value

> "Turn your camera into customizable ASCII art—fast, smooth, and expressive."

🔑 CORE FEATURES (UPDATED)

🎥 Live ASCII Camera

- Real-time rendering
- Monospace output
- LIVE indicator (pulse)

🎨 Customization System

🎭 *Presets*

- Cyberpunk (default)
- Classic
- Blocky
- Minimal

Each preset controls:

- character set
- contrast
- density defaults
- palette (initial)

🎨 *Color System (NEW)*

- Color toggle (on/off)
- If ON → show **RGB color picker**
- User selects any color from palette
- Selected color overrides preset palette

👉 Behavior:

- Applies single-tone color across ASCII
- Future-ready for gradient (optional)

🔤 *Character System (UPDATED)*

- Default preset character sets
- User can:
  - choose **single custom character only**
  - OR select from predefined sets

👉 Restriction:

- Only 1 character allowed (performance + consistency)

🎛 Controls

- Density slider
- Resolution slider
- Color toggle + RGB picker
- Invert toggle

📸 Capture System

- Flash (100ms) → shimmer (250ms)

- Freeze frame
- Camera pauses
- “This is captured (not live)”

📋 Export

- Copy ASCII:
  - compact
  - full
- Download image

⚙️ Performance System

- FPS-based adaptive scaling:
  - <20 FPS (5–7s) → degrade
  - ≥30 FPS (5s) → stable
- Step-based resolution scaling
- Cooldown (4–5s)
- Lower + upper bounds

🧠 Optimization System

- First-time calibration (optional)
- “Optimize” vs “Default”
- Manual re-run in settings

💾 Persistence

Stored in localStorage:

- last preset
- density
- resolution
- color selection
- custom character

- optimization status

🧾 Capture History

- Last 2 captures
- Silent replacement

📱 Mobile Support

- Bottom sheet controls
- Thumb-friendly capture
- Lower default resolution

🎨 2. PRODUCT DESIGN (UPDATED)

🧱 Layout

🖥 Desktop

```
[ Camera Preview (Main) ]
[ Collapsible Right Panel ]
```

📱 Mobile

```
[ Camera Preview ]
[ Capture Button ]
[ Bottom Sheet Controls ]
```

🔘 Key UI Elements

🎥 Camera Panel

- ASCII preview
- LIVE pulse indicator

🎯 Primary Action

- Large **Capture button**

⚙️ Controls Panel (COLLAPSIBLE ✅)

- Can expand / collapse
- Does NOT block camera permanently

🎛️ Control Sections

🎨 STYLE

- Presets
- Hover preview (precomputed thumbnails)
- Auto-switch to “Custom” on change

🌈 LOOK (UPDATED)

- Color toggle
- RGB color picker (visible when ON)
- Invert toggle

🔤 CHARACTER (NEW SECTION)

- Preset characters
- Single-character input
- Live preview updates

⚙️ QUALITY

- Density slider
- Resolution slider

📊 STATUS

- FPS meter
- Performance mode label

⚙️ Settings Menu

- Opened via **settings icon**
- Modal or panel

Contains:

- Run Performance Optimization
- Reset to default
- Sound toggle
- “Optimized X minutes ago”

🔄 3. UX FLOW (UPDATED)

🟢 FIRST-TIME FLOW

Step 1: Camera Permission

- If denied:

  “Enable camera from browser settings”

Step 2: Choice

- Optimize (primary)
- Default (secondary)

🟡 OPTIMIZATION PATH

- Blurred background
- “Optimizing…”
- Progress indicator
- Checkmark
- “Optimized for your device”
- Enter LIVE

🌐 DEFAULT PATH

- "Using default settings"
- Hint:

  "You can optimize later in settings"

⬤ LIVE STATE

User can:

- Adjust controls
- Change color
- Select character
- Switch presets

📸 CAPTURE FLOW

1. Click capture
2. Flash → shimmer
3. Freeze frame
4. Camera pauses

📄 RESULT STATE

- ASCII output
- Copy / download
- Retake

🔄 RETURN FLOW

- Retake → back to live

⚙ SETTINGS FLOW

- Open via icon
- Run optimization

- Reset

⚙️ 4. TECH STACK + ARCHITECTURE (UPDATED)

🧱 STACK

🎨 UI

- React
- CSS / Tailwind

🌐 Browser Layer

- Canvas API
- getUserMedia
- requestAnimationFrame
- localStorage

⚙️ Core Engine

- C++ → Emscripten

🔄 FRAME PIPELINE

```
Video → Canvas → Pixel Buffer → WASM → ASCII → Render
```

🧠 C++ RESPONSIBILITIES (UPDATED)

- ASCII conversion
- Brightness mapping
- Density + resolution
- Presets

- Performance system

NEW:

- Apply custom color
- Apply custom character

🎨 COLOR SYSTEM (IMPLEMENTATION)

If user selects color:

```
C++ColoruserColor;
booluseCustomColor;
```

Logic:

```
C++if (useCustomColor) {
applySingleColor(userColor);
} else {
usePresetPalette();
}
```

🔤 CHARACTER SYSTEM

```
C++charselectedChar;
booluseCustomChar;
```

Logic:

```
C++if (useCustomChar) {
returnselectedChar;
} else {
returnmapChar(brightness, charset);
}
```

🔌 JS ↔ WASM

- Pass:
  - pixel buffer
  - settings (density, color, char, preset)
- Receive:

- ASCII buffer

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⚡ PERFORMANCE SYSTEM

- FPS averaged
- Degrade / stabilize
- Step-based scaling